

**BANARAS HINDU UNIVERSITY**  
B.Sc. (Hons.) Semester VI Examination, 2013-14

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**Botany**

**Paper: BOB-601 — Plant Metabolism, Biochemistry and Biotechnology**

Time: Three hours

Full Marks: 70

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*Note: Attempt five questions in all, selecting two questions each from Section-A and B. Question No. 9 is compulsory. The figures in the right-hand margin indicate marks.*

**SECTION – A**

**Question 1.** What are co-enzymes? How do the allosteric enzymes differ from the enzymes following Michaelis-Menten kinetics? [15]

**Question 2.** Describe the pathway of sucrose biosynthesis in plants. [15]

**Question 3.** Write critical notes on any two of the following: [7.5 × 2 = 15]

- (a) Ammonia assimilation
- (b) Mechanism of O<sub>2</sub> regulation during N<sub>2</sub> fixation
- (c) Biosynthesis of cellulose

**Question 4.** Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Nitrogenase enzyme
- (b) Activation and assimilation of phosphorus
- (c) Isoenzymes

**SECTION – B**

**Question 5.** What are auxins? Describe their biosynthetic pathways and mode of action. [15]

**Question 6.** What is DNA replication? Describe the process of DNA replication and its control mechanisms. [15]

**Question 7.** Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Cloning vectors
- (b) B-DNA
- (c) Recombinant DNA technology

**Question 8.** Briefly describe any two of the following: [7.5 × 2 = 15]

- (a) Transcription
- (b) Absciscic acid
- (c) Post-translational modification of proteins

### SECTION – C

**Question 9.** Fill in the blanks or answer the following in not more than 50 words each:  
[1 × 10 = 10]

- (a) What are energy-rich phosphorus compounds?
- (b) Cell division can be stimulated by \_\_\_\_\_ hormone.
- (c) Define prosthetic groups.
- (d) \_\_\_\_\_ is the site of lipid synthesis in plants.
- (e) EF-Ts is an \_\_\_\_\_ factor.
- (f) What is the role of topoisomerase?
- (g) \_\_\_\_\_ gene encodes DNA polymerase I.
- (h) The number of base pair per turn of A-DNA helix is \_\_\_\_\_.
- (i) What is a shuttle vector?
- (j) \_\_\_\_\_ is an initiation codon.

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